USDA National Institute of Food and Agriculture
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TEXAS A&M AGRILIFE RESEARCH

CANVAS
Where Crop, Agronomic, Environmental, and Soil Sciences Connect





Integrating Electrical, Thermal, and Mechanical Tactics for Inter- and Intra-Row Weed Management in Organic Grain Sorghum

ASA | CSSA | SSSA



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Why It Matters : Contributing to Five UNSDGs



Zero Hunger

Drought-hardy grain helps feed a warming world.



Good Health

Cleaner grain and water, with no synthetic chemicals.



Responsible Production

Herbicide-free weed control cuts chemical inputs.



Climate Action

Drought-tolerant sorghum stays productive in extreme weather.



Life on Land

Fewer inputs protect soil life and slow resistance.



Sustainable, **chemical-free weed control** is not only good agronomy, it advances a **global agenda** for **food, health, climate, and land**.

SUSTAINABLE DEVELOPMENT GOALS

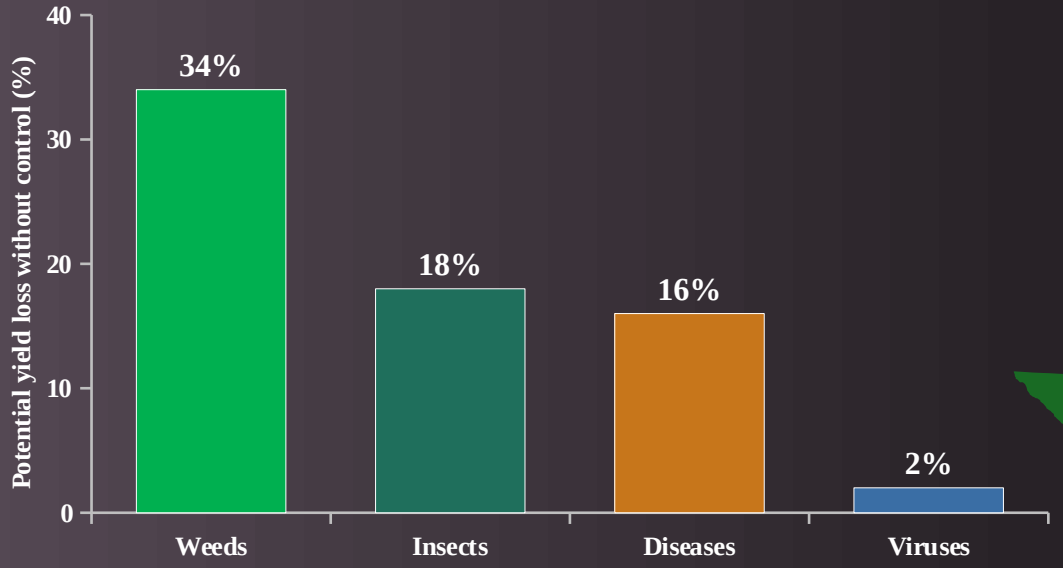


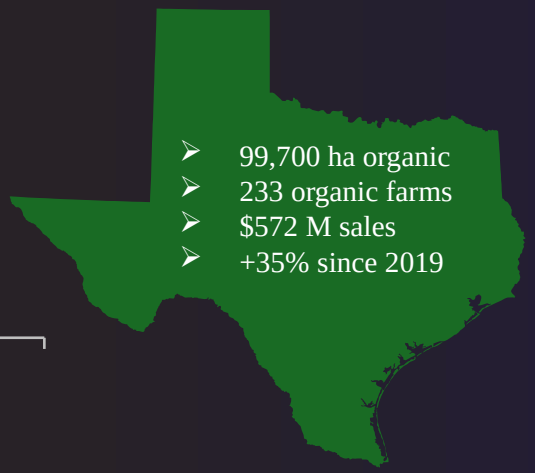
Organic Agriculture: Global Growth and the Texas Context

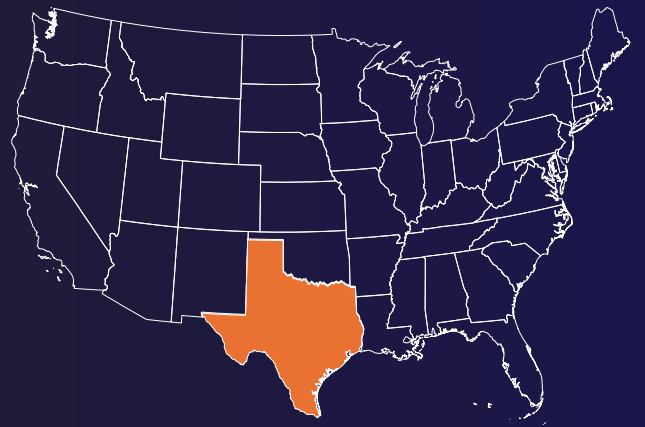
- Organic is booming: $11.2B in U.S. sales; Texas up 35% since 2019.
- Weeds are the **No. 1** threat — up to 34% yield loss.
- In sorghum, Palmer amaranth alone can cut yield 32–63%.
- Even where herbicides are allowed, they're failing — resistant sorghum weed cases rose from 2 (1993) to 16 (2023).



World: 98.9 M ha farmland • 183 countries • $145 B retail







USA: 2.0 M ha organic • 17,445 organic farms • $11.2 B sales

Source: *Oerke, E.-C. (2006).*

Two Zones, One Blind Spot

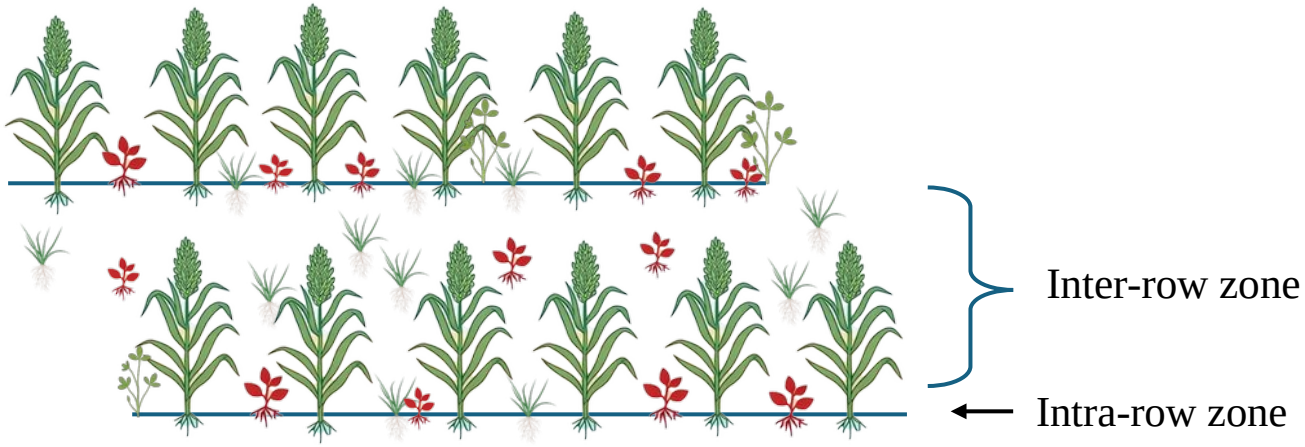
- Weeds grow in two places: between the rows (inter-row) and within the row (intra-row).
- Between the rows is the easy part, cultivation, flaming and EWC can reach it.
- Within the row is the hard zone; right next to the crop, no herbicide, and hand-weeding is the costly fallback.
- The weeds inside the row are the real problem. This study asks how to control that in-row zone using non-chemical tools alone.



(a) Weed sampling in intra-row zone



(b) Weed sampling in inter-row zone





No single tool controls both zones; so we combine electric, flame, and cultivation.

The Toolkit: Thermal, Mechanical & Electrical Weed Control



THERMAL

Propane Flaming

Intense heat ruptures weed cells

Zone: Both zones



MECHANICAL

Mechanical Cultivation

Blades uproot and bury weeds between the rows.

Zone: Between rows only



ELECTRICAL

Electrical Weeding (EWC)

High-voltage current kills on contact

Zone: Between rows only

Each tool covers a different zone with the different MOA — combined, they replace the herbicide across the whole field.

Objectives & Hypothesis

1. **EVALUATE : Single vs. integrated tactics**
   Electrical (EWC), thermal (flaming) and mechanical (cultivation), applied alone and in integrated programs.

2. **QUANTIFY : Weed control efficacy (WCE)**
   WCE (%) by spatial zone (inter- vs. intra-row) and species group (grasses vs. broadleaves).

3. **ASSESS : Crop safety**
   Visual crop injury (%) and canopy recovery across days after treatment (DAT).

4. **OPTIMIZE : Selectivity**
   Identify the program maximizing WCE with minimal crop injury (combined selectivity index).

HYPOTHESIS

Integrated, multi-tactic programs targeting both zones will achieve higher weed control efficacy than any single tactic *— while keeping crop injury minimum and within acceptable limits.*

Aim: a non-chemical program that maximizes weed control efficacy while preserving crop safety.

Materials & Methods



Aerial RGB image of all 32 plots [Flowering Stage].

8 Treatments | **4** Reps | **32** Plots

STUDY SITE & DESIGN

Location: College Station, TX
Crop: Organic grain sorghum
Season: 2026 **Plot:** 20 × 35 ft
Design: RCBD (8 × 4 = 32 plots)

DATA COLLECTED & ANALYSIS

- Weed density → WCE (%), by zone
- Crop injury (%) at 1, 3, 7, 14 & 28 DAT
- Canopy / vigor from RGB imagery
- ANOVA + Fisher's protected LSD, α = 0.05

Treatment	T1	T2	T3	T4	T5	T6	T7	T8
Inter-row	EWC	★	EWC	Flaming	Flaming	Control (★)	Cultivation	Cultivation
Intra-row	★	Flaming	Flaming	★	Flaming	Control (★)	★	Flaming

Design: eight non-chemical weed-control programs, replicated four times and imaged across the season.

Treatments in Action : Operating Parameters



EWC (Electrical)

- **255 V · 18.25 A**
- Power = 4.65 kW
- Resistance = 13.97 Ω
- Travel speed = 2 mph
- No soil disturbance



Cultivation (Mechanical)

- Travel speed = 2 mph
- Inter-row sweeps
- Removes between-row weeds
- In-row weeds remain



Inter-row Flaming

- Propane = **30 psi** (2.2 bar)
- Travel speed = 2 mph
- Burners parallel to row **(180°)** tilted down
- Hood fitted, **not used**



Intra-row Flaming

- Propane = **30 psi** (2.2 bar)
- Travel speed = 2 mph
- Cross-burner setup
- Angled **30–45°** to crop row
- Offset **10–15 cm** from row

Treatment Gallery



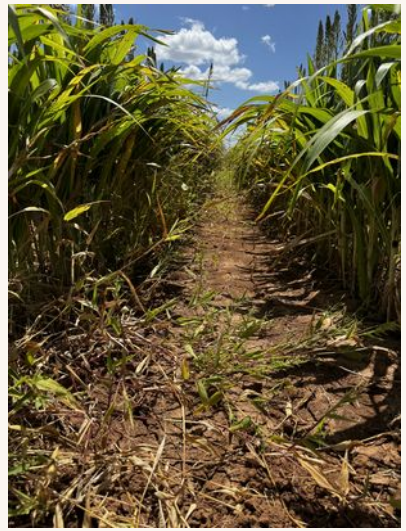
(a) T1
Inter-row EWC only



(b) T2
Intra-row flaming only



(c) T3
Inter-row EWC + Intra-row flaming



(d) T4
Inter-row flaming only



(e) T5
Inter-row flaming + Intra-row flaming



(f) T6
Untreated weedy check (control)

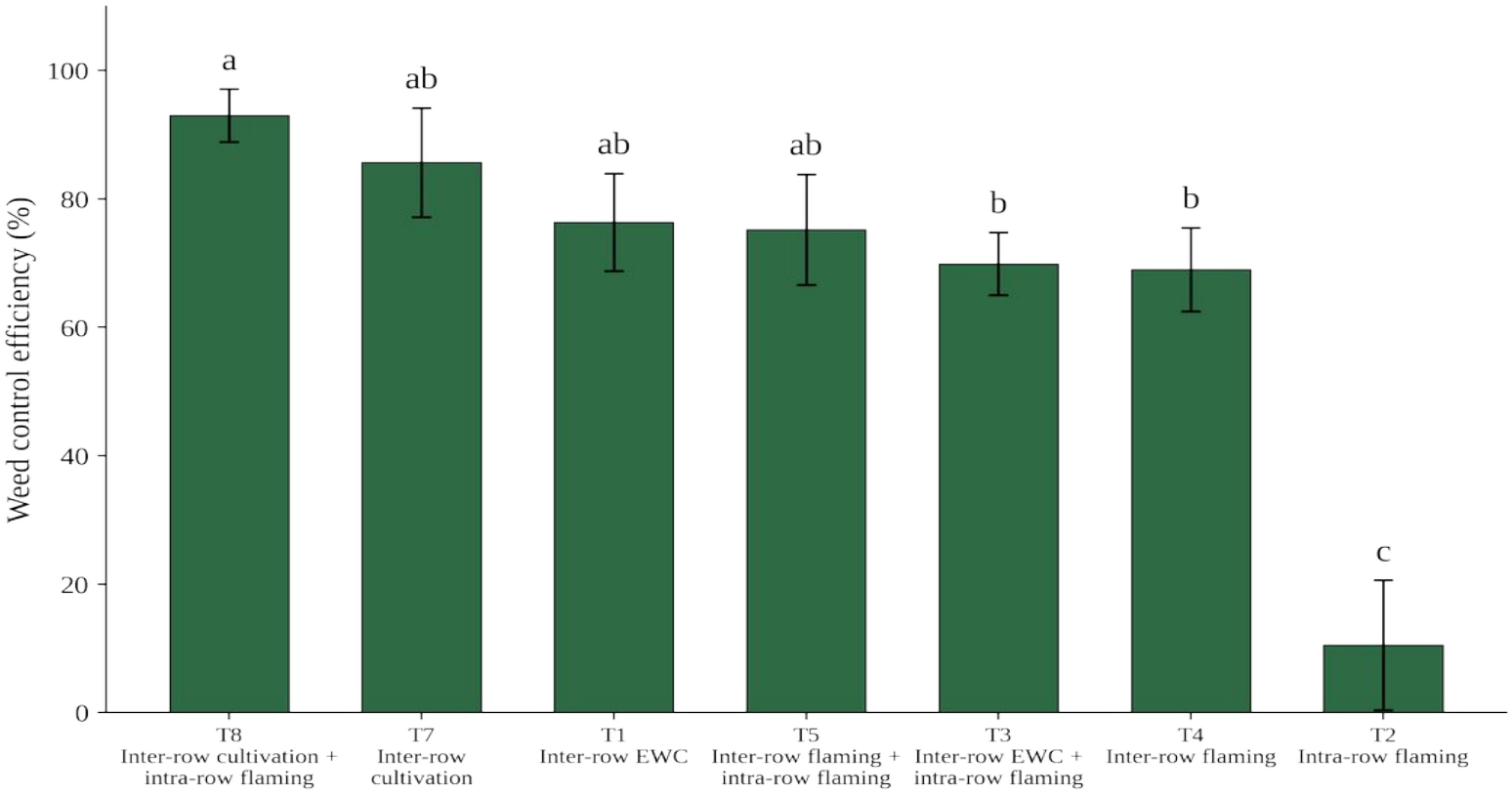


(g) T7
Inter-row cultivation only



(h) T8
Cultivation + Intra-row flaming

Results: Inter-row Weed Control Efficiency



93.0%

T8 · Inter-row cultivation + intra-row flaming

highest inter-row control (group a)

85.7%

T7 Cultivation only

best single pass (ab)

10.5% ↓

T2 Intra-row flaming only

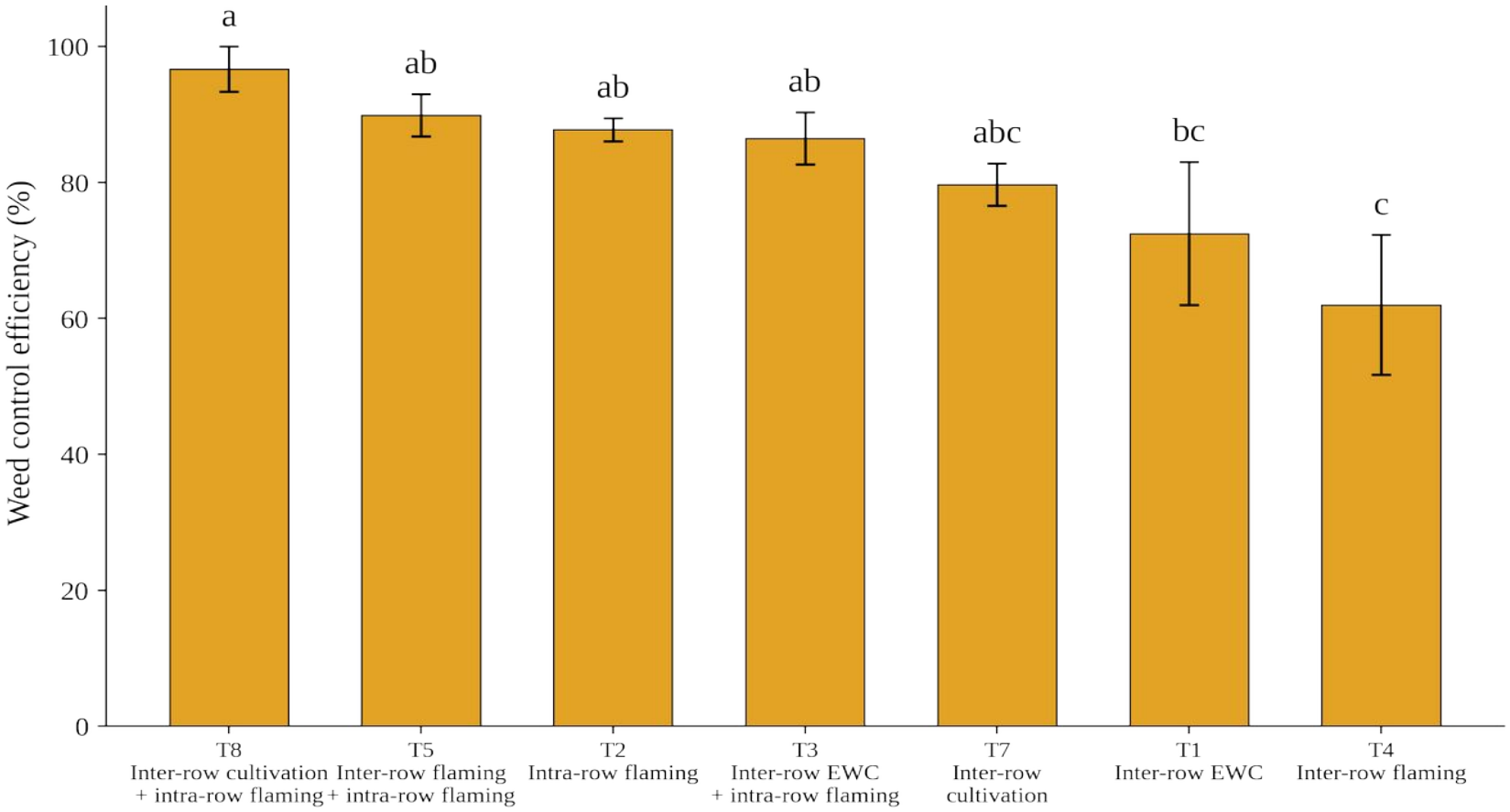
no inter-row effect (c)

Take-home: each inter-row pass worked; **cultivation best**; intra-flame alone did nothing between the rows.

Key Findings

1. Inter row weed management achieved **69–93%** inter-row control, led by **T8 (93.0%, a)** and inter-row cultivation **T7 (85.7%, ab)**.
2. Intra-row flaming alone (T2) reached only **10.5% (c)**, statistically equal to the untreated check (Control).
3. Cultivation produced the **lowest inter-row grass density** (T7, T8), performing best on the tougher grasses.

Results: Intra-row Weed Control Efficiency



96.7%

T8 · Inter-row cultivation + intra-row flaming

highest in-row control (group a)

87.8%

T2 Intra-row flaming only

matches the full programs (ab)

62.0%

T4 Inter-row flaming only

weakest in the crop row (c)

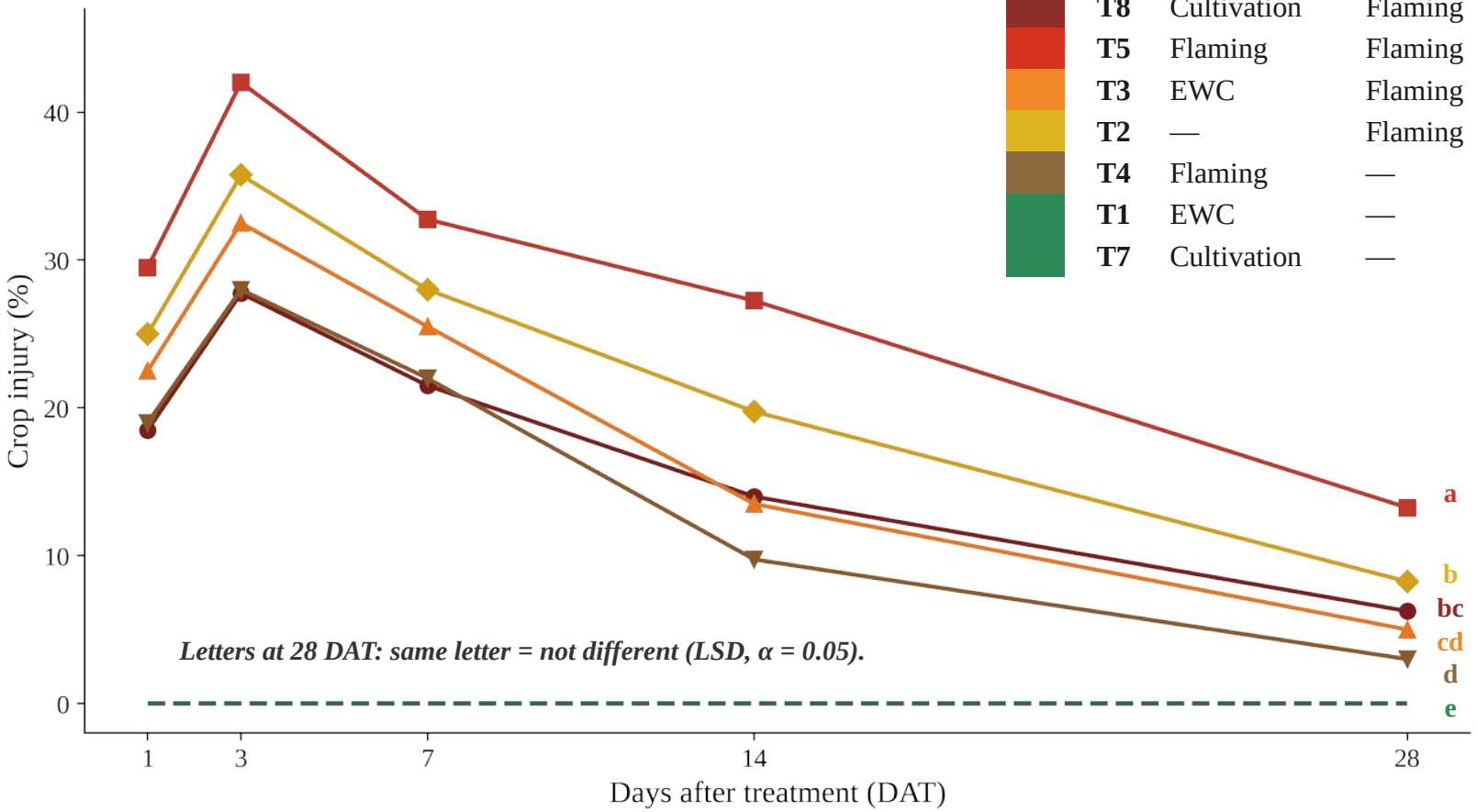
Take-home: only an **intra-row pass** reaches the crop line.

Key Findings

1. Intra-row flaming drove in-row control, led by **T8 (96.7%, a)**, with **T5, T2, T3** on par at **86–90% (ab)**.
2. Inter-row-only tools were weakest here — inter-row flaming **T4 (62.0%, c)** and EWC **T1 (72.5%, bc)** can't reach the crop line.
3. Flaming nearly eliminated broadleaves (**waterhemp, Palmer 89–100%**), while grasses like **barnyardgrass** regrew.

Crop Safety: Injury & Recovery

Trt	Inter-row	Intra-row
T8	Cultivation	Flaming
T5	Flaming	Flaming
T3	EWC	Flaming
T2	—	Flaming
T4	Flaming	—
T1	EWC	—
T7	Cultivation	—





Crop injury from intra-row flaming

Key Findings

1. Crop injury was transient, peaking at **~3 DAT** and declining steadily to **<15% by 28 DAT** in every treatment.
2. **EWC and cultivation caused no injury (0%)**, while flaming was the only source of damage, highest in **T5 (42% → 13%)** and lowest in **T4 (28% → 3%)**.
3. The best weed-control treatment **T8** stayed crop-safe, with injury recovering from **28% to ~6%** by 28 DAT.

Conclusions

1. Only **T8, T7** and **T3** reached the **ideal zone**, achieving high weed control together with high crop safety (**at least 70% control and no more than 10% residual injury**).

2. **T8** was the standout treatment, combining about **93% inter-row and 97% intra-row control** with about **94% crop safety**, leaving only about 6% lasting injury.

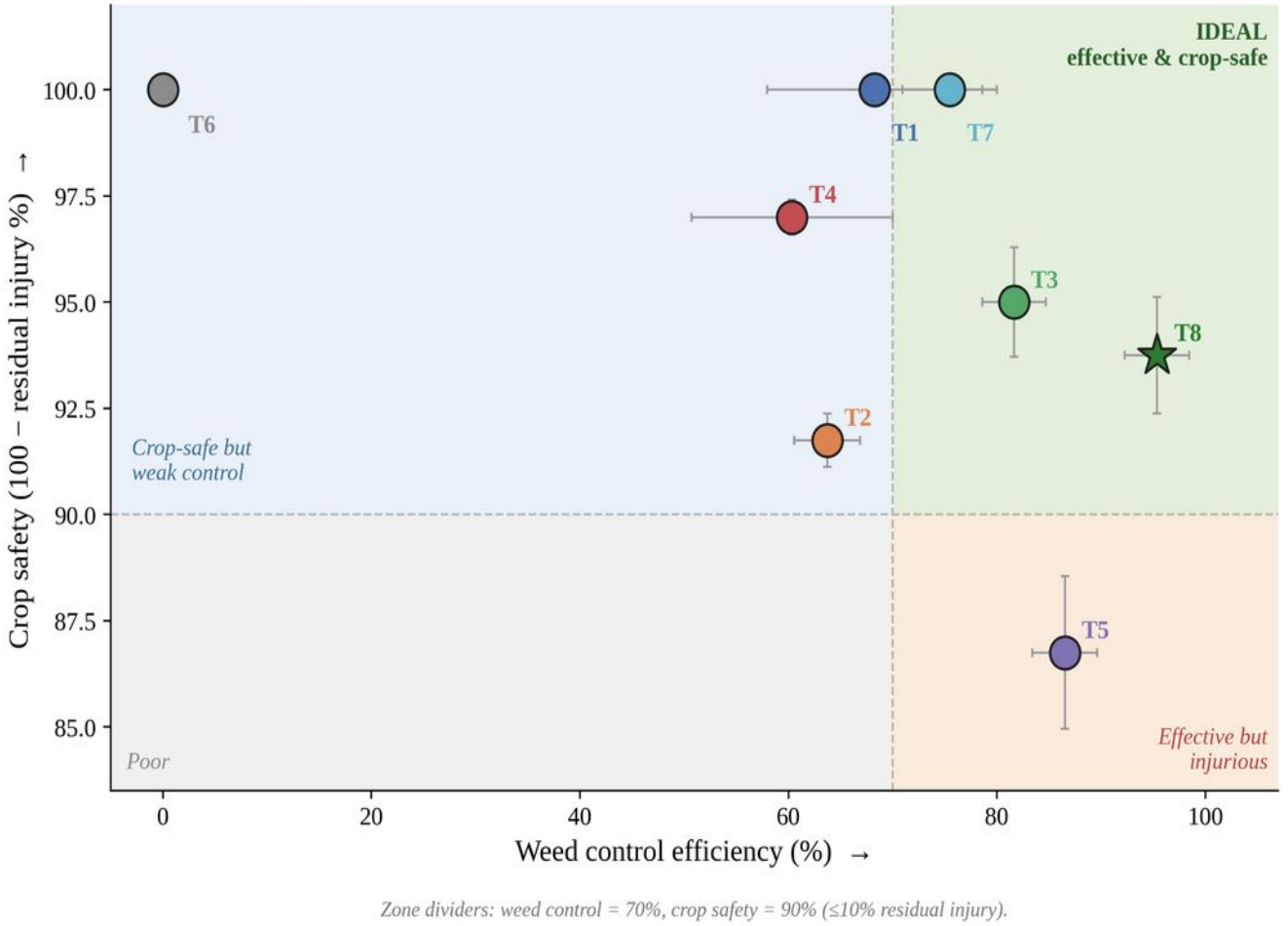
BEST PROGRAM

T8 — *Inter-row cultivation + Intra-row flaming*

93% inter · **97%** intra · **crop-safe**

FUTURE WORK

1. **Yield:** test whether any treatment carries a yield penalty.
2. **Canopy recovery:** track crop recovery with hyperspectral data.
3. **Weed-free check:** add a hand-weeded or chemical plot to compare yield.
4. **Cotton:** repeat the study in cotton-sorghum cropping systems.



Zone dividers: weed control = 70%, crop safety = 90% (≤10% residual injury).

Recommendation:

Cultivation + intra-row flaming (T8) gives **herbicide-free, crop-safe control**, next step — improving control of flame-tolerant grasses (barnyardgrass).

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